

Physical AI Predictive Maintenance: Monitoring Industrial Motors Using Distributed Edge Sensors

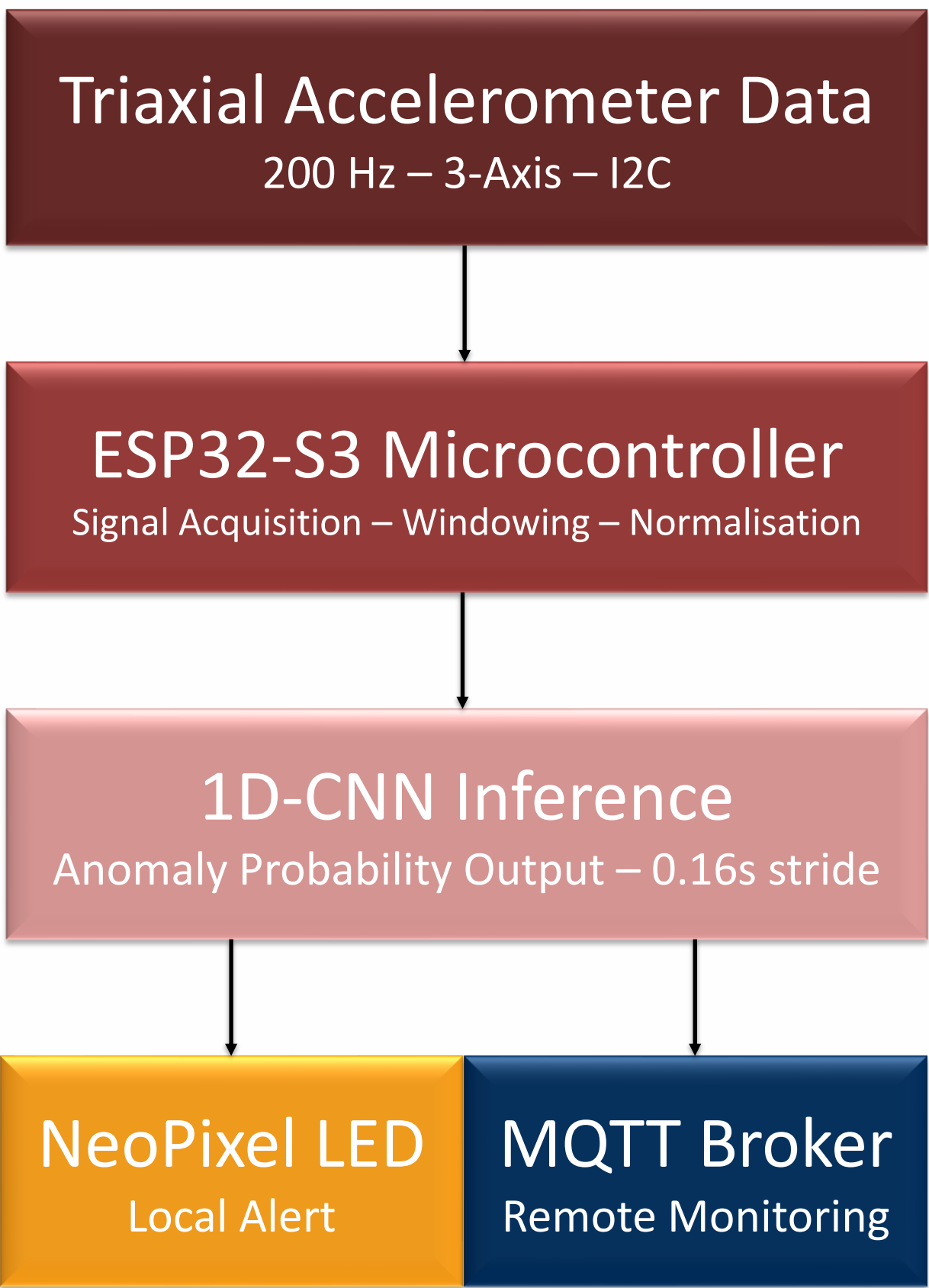
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SUMMARY

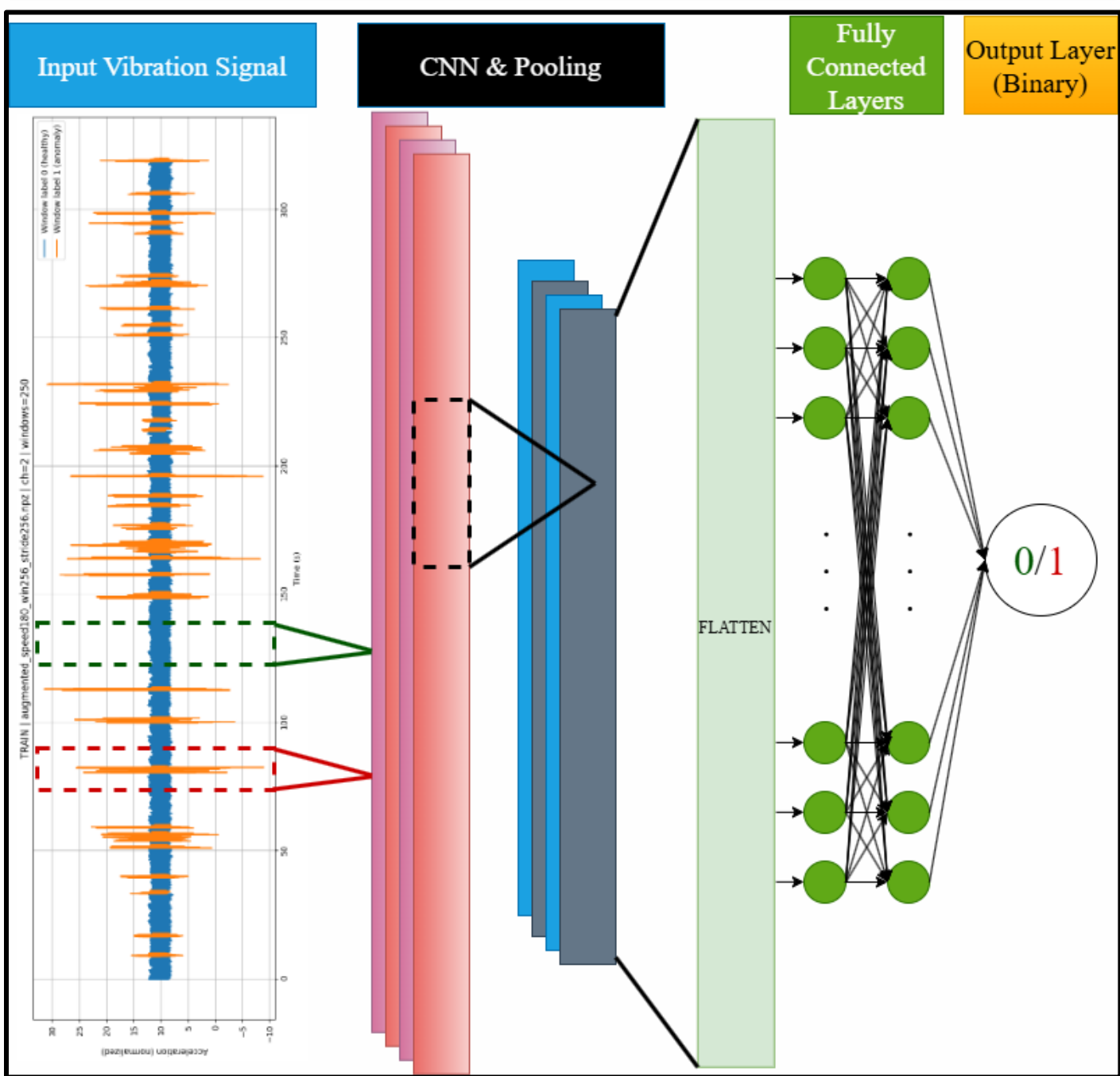
- Unplanned downtime costs the world's largest industrial organisations approximately 11% of their annual revenue.
- Vibration data collected via triaxial ADXL345 accelerometer at 200 Hz on a NEMA17 stepper motor rig.
- Iterative development process: regression baseline -> 1D-CNN -> data augmentation -> multi-speed CNN training.
- Example driven anomaly augmentation used to overcome fault data scarcity.
- Quantised 1D-CNN deployed on ESP32-S3: 145 kB model, 11.6 kB RAM at runtime.
- Electrical and Mechanical fault detections: 87.5% - 99.6% across seen and unseen operating speeds.
- Remote monitoring via lightweight MQTT (63 bytes/message).
- Inference every 0.16 seconds.

SYSTEM ARCHITECTURE

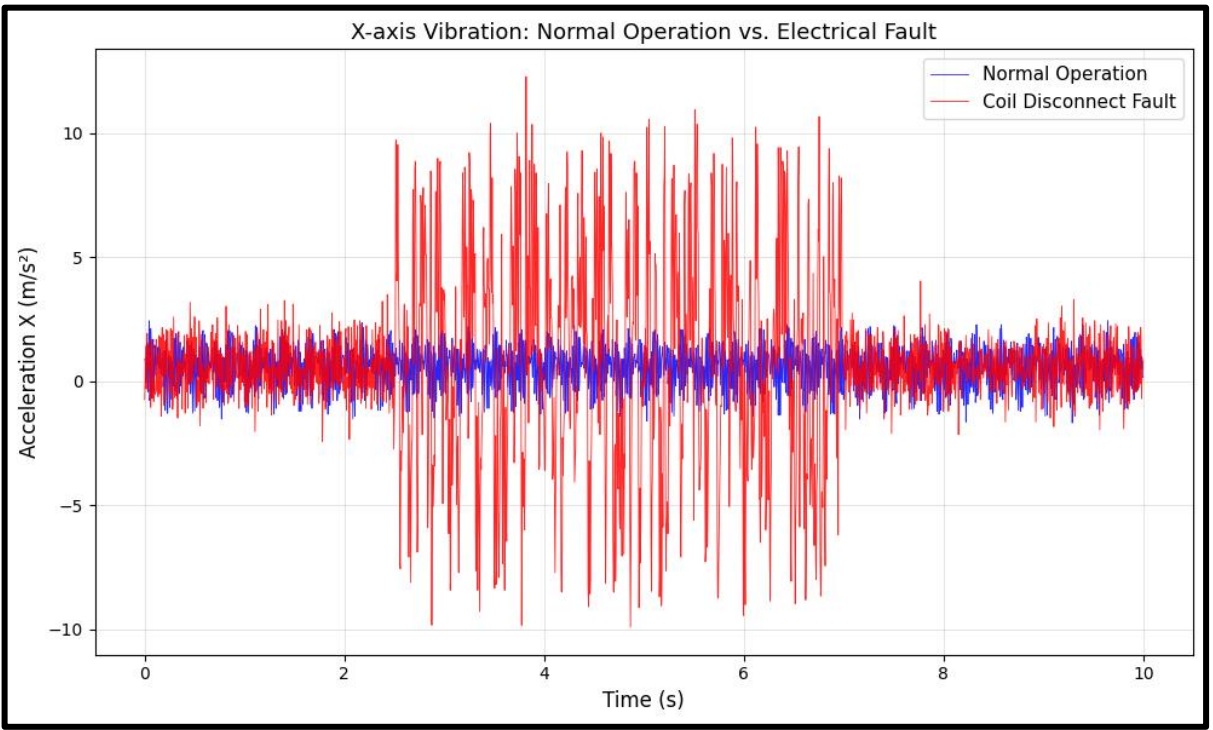


- Fully standalone Edge AI monitoring device.
- No cloud dependencies.
- Sensing, inference, IoT communication run concurrently on low-cost microcontroller.

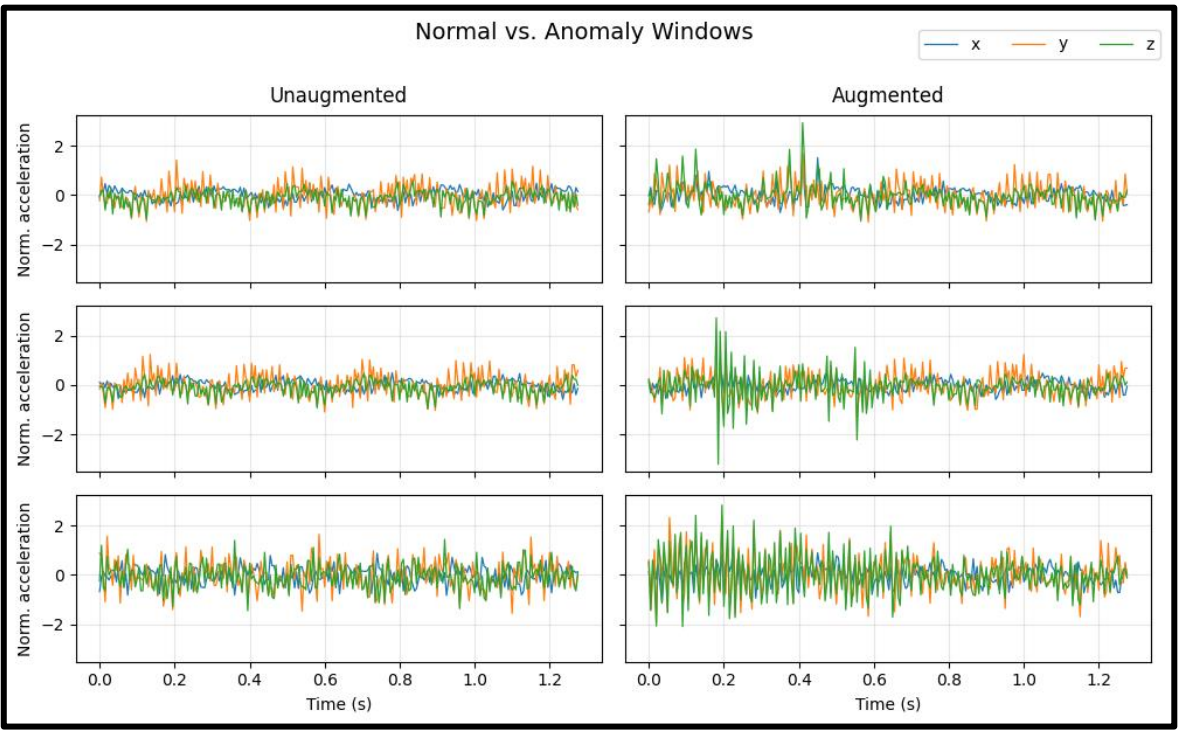
1D-CNN ARCHITECTURE



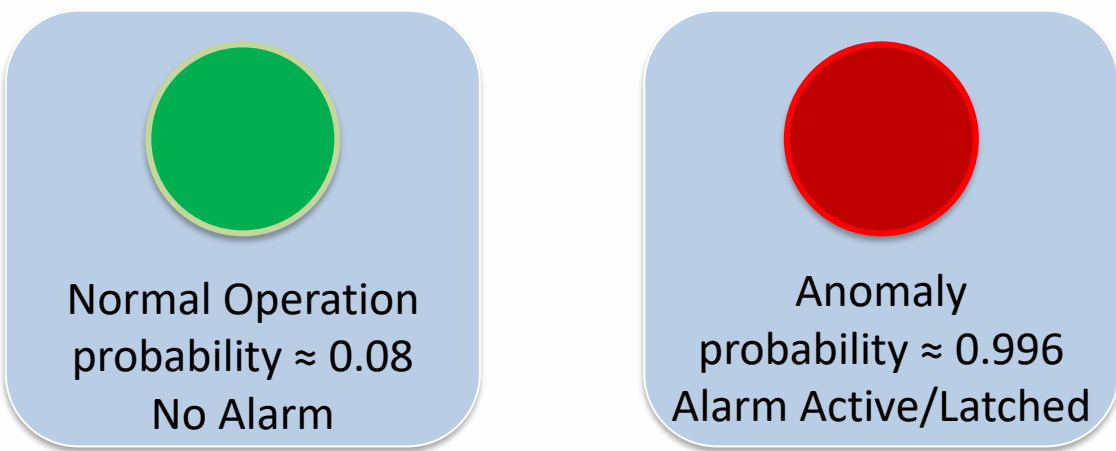
VIBRATION SIGNALS



ANOMALY AUGMENTATION



IOT MONITORING

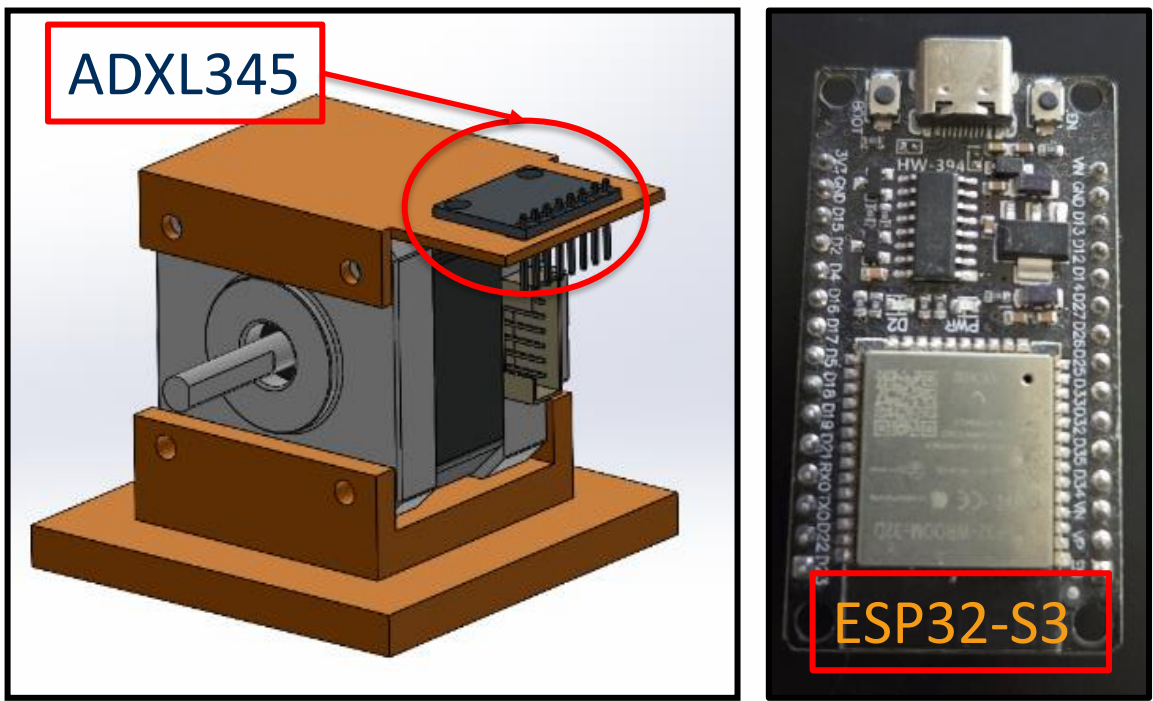


- MQTT Payload: 63 bytes
- Alarm latches until manual reset by operator

KEY RESULTS

87.5%-99.6% Anomaly probabilities for induced faults	145 kB Quantised model flash footprint
11.6 kB Runtime RAM (Tensor Arena)	~15 min Training data required per speed
Trained on: 3 Speeds: 29.3/39.1/52.1 RPM	Tested on: Range: 11.7 - 156.25RPM

EXPERIMENTAL RIG



A fully independent Edge AI PdM system was validated across multiple motor speeds, detecting unseen resonance and electrical faults on trained and untrained motor speeds with >87.5% confidence. System achieved proof of concept results using only 145 kB flash and 11.6 kB RAM on a ESP32-S3.